

IP LEET 2018

QUESTION BOOKLET SERIES : A

Max. Marks : 600

No. of Questions : 150

Time : 150 min

Instructions :

- Write your roll No. on the Question Booklet and also on the OMR Answer Sheet in the space provided and no where else.
- Enter the Code No. of Question Booklet on the OMR answer Sheet. Darken the corresponding bubbles with Black Ball Point/ Black Gel Pen.
- Do not make any identification mark on the Answer Sheet or Question Booklets.
- To open the Question Booklet remove the seal gently when asked to do so.
- Please check that this Question Booklet contains 150 questions. In case of any discrepancy, inform the Assistant Superintendent within 10 minutes of the start of test.
- Each question has four alternative answer (A, B, C, D) of which only one is correct. For each question, darken only one bubble (A or B or C or D), whichever you think is the correct answer, on the Answer Sheet with Black Point/Black Gel Pen. There shall be negative marking for wrong answers.
- If you do not want to answer a question, leave all the bubbles corresponding to that question blank in the Answer Sheet. No marks will be deducted in such cases.
- Darken the bubbles in the OMR Answer Sheet according to the Serial No. of the questions given in the Question Booklet.
- If you want to change an already marked answer, erase the shade in the darkened bubble completely.
- For rough work only the blank sheet at the end of the Question Booklet be used.
- The Answer Sheet is designed for computer evaluation. Therefore, if you do not follow the instructions given on the Answer Sheet, it may make evaluation by the computer difficult. Any resultant loss to the candidate on the above account, i.e. not following the instructions completely, shall be of the candidate only.
- After the test, hand over the Question Booklet and the Answer Sheet to the Assistant Superintendent on duty.
- In no case the Answer Sheet, the Question Booklet, or its part or any material copied/ noted from this Booklet is to be taken out of the examination hall. Any candidate found doing so would be expelled from the examination.
- A candidate who creates disturbance of any kind or changes his/her seat or is found in possession of any paper possibly of any assistance or found given or receiving assistance or found using any other unfair means during the examination will be expelled from the examination by the Centre Superintendent/Observer whose decision shall be final.
- Telecommunication Equipment such as mobile phones, pager, cellular phone, wireless set, scanner, camera or any electronic gadget etc., is not permitted inside the examination hall. Use of calculators is not allowed.
- The candidates will not be allowed to leave the Examination Hall/Room before the expiry of the allotted time.

- The ratio of static friction to dynamic friction is always
(A) Equal to one (B) Less than one
(C) Greater than one (D) None of these.
- Moment of inertia of a triangular section of base (b) and height (h) about an axis through its base, is
(A) $bh^3/4$ (B) $bh^3/8$
(C) $bh^3/12$ (D) $bh^3/36$
- Fig. Q3 below shows the three coplanar forces P , Q and R acting at a point O . If these forces are in equilibrium, then

$$\begin{aligned} \text{(A)} \quad & \frac{P}{\sin \beta} = \frac{Q}{\sin \alpha} = \frac{R}{\sin \gamma} \\ \text{(B)} \quad & \frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \\ \text{(C)} \quad & \frac{P}{\sin \gamma} = \frac{Q}{\sin \alpha} = \frac{R}{\sin \beta} \\ \text{(D)} \quad & \frac{P}{\sin \alpha} = \frac{Q}{\sin \gamma} = \frac{R}{\sin \beta} \end{aligned}$$

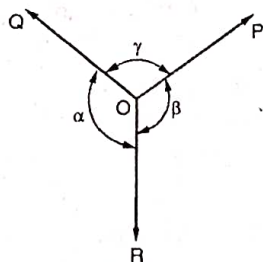


Fig. Q3.

- The angle of inclination of the plane at which body begins to move down the plane, is called
(A) Angle of friction (B) Angle of repose
(C) Angle of projection (D) None of these.
- The term 'Centroid' refers to
(A) The same as centre of gravity.
(B) The point of suspension.
(C) The point of application of the resultant of all the

forces tending to cause a body to rotate about a certain axis.

(D) None of the above.

- If the masses of both the bodies, as shown in given Fig. Q6, are reduced to 50 percent, the tension in the spring will be

(A) Same (B) Half
(C) Double (D) None of these.

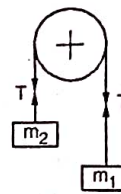


Fig. Q6.

- The moment of inertia of a thin rod of mass ' m ' and length ' l ' about an axis through its centre of gravity and perpendicular to its length is

(A) $\frac{ml^2}{4}$ (B) $\frac{ml^2}{6}$ (C) $\frac{ml^2}{8}$ (D) $\frac{ml^2}{12}$

- A rubber ball is dropped from a height of 2 m. If there is no loss of velocity after rebounding, the ball will rise to a height of

(A) 1 m (B) 2 m (C) 3 m (D) 4 m

- Which of the following is a scalar quantity?

(A) Force (B) Speed
(C) Velocity (D) Acceleration

- A weight of 1000 N can be lifted by an effort of 80 N. If the velocity ratio is 20, the machine is

(A) Reversible (B) Non-reversible
(C) Ideal (D) None of these.

- The resultant of two forces P and Q (such that $P > Q$) acting along the same straight line, but in opposite direction, is given by

(A) $P + Q$ (B) $P - Q$
(C) P/Q (D) Q/P

- Which of the following is an example of lever of first order?

(A) Arm of man (B) Pair of scissors
(C) Pair of clinical tongs (D) All of the above.

- A block of mass m_1 , placed on an inclined smooth plane is connected by a light string passing over a smooth pulley to mass m_2 , which moves vertically downward as shown in the Fig. Q13. The tension in the string is

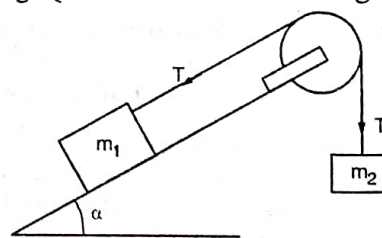


Fig. Q13.

(A) m_1/m_2 (B) $m_1 g \sin \alpha$
(C) $\frac{m_1 m_2}{m_1 + m_2}$ (D) $\frac{m_1 m_2 g (1 + \sin \alpha)}{(m_1 + m_2)}$

- The velocity of a body on reaching the ground from a height, h , is

(A) $2\sqrt{gh}$ (B) $\sqrt{gh}$
(C) $\sqrt{2gh}$ (D) $2g\sqrt{h}$

15. In order to double the period of simple pendulum, the length of string should be
(A) Halved (B) Doubled
(C) Quadrupled (D) Same.
16. The three forces of 100 N, 200 N and 300 N have their lines of action parallel to each other but act in the opposite directions. These forces are known as
(A) Coplanar concurrent forces
(B) Coplanar non-concurrent forces
(C) Like parallel forces
(D) Unlike parallel forces.
17. Moment of inertia of a hollow circular section, as shown in the Fig. Q17, about an axis perpendicular to the section, is times than that about X-X axis.
- (A) Two
(B) Same
(C) Half
(D) None of these.
18. In actual machines,
(A) Mechanical advantages is greater than velocity ratio
(B) Mechanical advantages is equal to velocity ratio
(C) Mechanical advantages is less than velocity ratio
(D) Mechanical advantages is unity.
19. The centre of gravity of a T-section 100 mm × 150 mm × 50 mm from its bottom is
(A) 50 mm (B) 75 mm
(C) 87.5 mm (D) 125 mm.
20. The length of second's pendulum is
(A) 94.9 cm (B) 99.4 cm
(C) 100 cm (D) 101 cm
21. The angle between two forces when the resultant is maximum and minimum respectively are
(A) 0° and 180° (B) 180° and 0°
(C) 90° and 180° (D) 90° and 0°
22. The velocity ratio of a first system of pulleys with 4 pulleys is
(A) 4 (B) 8 (C) 16 (D) 20
23. The unit of angular velocity is
(A) m/min (B) rad/s
(C) revolutions/min (D) Both (B) and (C).
24. The force induced in the string BC due to load 'W' as shown in Fig. Q24, is

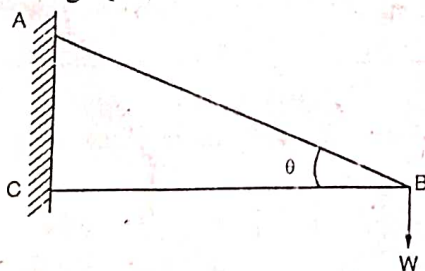


Fig. Q24.

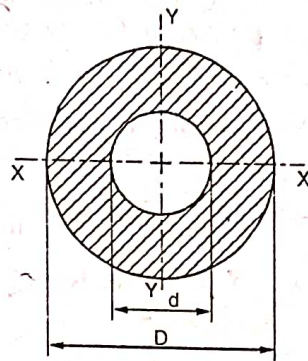


Fig. Q17.

- (A) $W \sin \theta$ (B) $W \cos \theta$ (C) $W \tan \theta$ (D) $W \cot \theta$
25. If three forces acting in different planes can be represented by a triangle, these will be in
(A) Non-equilibrium (B) Partial equilibrium
(C) Full equilibrium (D) Unpredictable
26. The horizontal range of projectile (R) is given by
(A) $R = u^2 \cos 2\alpha/g$ (B) $R = u^2 \sin 2\alpha/g$
(C) $R = u^2 \cos \alpha/g$ (D) $R = u^2 \sin \alpha/g$.
27. The acceleration of a particle moving with simple harmonic motion is at the mean position.
(A) Zero (B) Minimum
(C) Maximum (D) None of the above.
28. Efficiency of screw jack is given by (where α = Helix angle, and ϕ = angle of friction)
(A) $\tan(\alpha + \phi)/\tan \alpha$ (B) $\tan \alpha / \tan(\alpha + \phi)$
(C) $\tan(\alpha - \phi)/\tan \alpha$ (D) None of these.
29. Centre of gravity of thin hollow cone lies on the axis at a height of
(A) one-fourth of the total height above base
(B) one-third of the total height above base
(C) one-half of the total height above base
(D) three-eighth of the total height above the base.
30. For a self locking machine, the efficiency must be
(A) equal to 50% (B) less than 50%
(C) greater than 50% (D) equal to 100%
31. If twice number of pulleys in a system is equal to velocity ratio, then it is a order system of pulleys.
(A) First (B) Second
(C) Third (D) None of these.
32. The ideal angle of banking provided on the curve roads depend upon
(A) Weight of vehicle (B) (Velocity)² of the vehicle
(C) Nature of road surface
(D) Coefficient of friction between the road and vehicle contact point.
33. The acceleration of a particle moving with simple harmonic motion, at any instant is given by
(A) $\omega \cdot y$ (B) $\omega^2 y$ (C) ω^2/y (D) $\omega^3 y$.
34. The range of projectile will be maximum for a given velocity of projectile, when the angle of projection (α) is
(A) $\beta/2$ (B) $30^\circ + \beta/2$
(C) $45^\circ + \beta/2$ (D) $60^\circ + \beta/2$
35. When the spring of a watch is wound, it will possess
(A) Strain energy (B) Kinetic energy
(C) Heat energy (D) Electrical energy
36. Two balls of equal mass and of perfectly elastic material are lying on the floor, one of the ball with velocity 'v' is made to strike the second ball. Both the balls after impact will move with a velocity.
(A) V (B) V/2 (C) V/4 (D) V/8
37. If the gravitational acceleration at any place is doubled, then weight of body will be
(A) $g/2$ (B) g (C) $\sqrt{2}g$ (D) 2g

38. When a body of mass moment of inertia ' I ' (about a given axis) is rotated about that axis with an angular velocity ω , then kinetic energy of rotation is
(A) $I\omega$ (B) $I\omega^2$ (C) $0.5 I\omega$ (D) $0.5 I\omega^2$

39. Magnetism at the centre of a bar magnet is
(A) Minimum (B) Maximum
(C) Zero (D) None of these.

40. Super conductors are substances which
(A) conduct electricity at low temperature
(B) offer high resistance to the flow of current
(C) offer no resistance to the flow of electricity
(D) conduct electricity at high temperature.

41. Optical fibre works on the principle of
(A) Refraction (B) Total internal refraction
(C) Scattering (D) Interference.

42. The values of resistance ' R ' for the given circuit as shown in Fig. Q42.

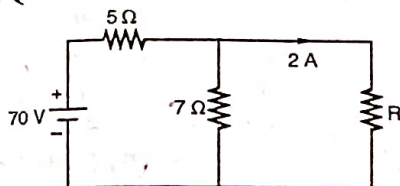


Fig. Q42.

- (A) 7Ω (B) 17.5Ω
(C) 10Ω (D) 12Ω
43. When pointer of an analog ohmmeter reads close to zero, the resistor being measured is
(A) open (B) reversed
(C) shortened (D) overheated.
44. If the resistance of wire is r ohms and wire is stretched to double of its length, then what will be its new resistance?
(A) r (B) $2r$
(C) $4r$ (D) $r/2$
45. A convex lens of focal length 40 cm is in constant with a concave lens of focal length 25 cm, the power of combination is
(A) -1.5 dioptres (B) -6.5 dioptres
(C) $+6.5$ dioptres (D) $+6.67$ dioptres
46. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. The number of spectral lines in the emission spectra will be
(A) 2 (B) 3
(C) 5 (D) 6.
47. A ray of light passes from vacuum into a medium of refraction index n . If the angle of incidence is twice the angle of refraction, the angle of incidence is
(A) $\cos^{-1}\left(\frac{n}{2}\right)$ (B) $\sin^{-1}\left(\frac{n}{2}\right)$
(C) $2\cos^{-1}\left(\frac{n}{2}\right)$ (D) $2\sin^{-1}\left(\frac{n}{2}\right)$
48. Two bodies of masses m and $4m$ are placed at a distance r . The gravitational potential at a point on the line joining them, where the gravitational field is zero, is

- (A) zero (B) $\frac{-4 Gm}{r}$
(C) $\frac{-6 Gm}{r}$ (D) $\frac{-9 Gm}{r}$

49. The electrostatic potential inside a charged spherical ball is given as $\phi = ar^2 + b$ where r is the distance from the centre ; a and b are constants. Then, the charge density inside the ball is

- (A) $-24\pi a \epsilon_0 r$ (B) $-6\pi a \epsilon_0 r$
(C) $-24\pi a \epsilon_0$ (D) $-6a\epsilon_0$

50. An object moving with a speed of 6.25 m/s, is decelerated at a rate given by

$$\frac{dV}{dt} = -2.5 \sqrt{V}$$

where V is the instantaneous speed. The time taken by object, to come to rest would be

- (A) 1 s (B) 2 s (C) 4 s (D) 8 s

51. Which of the following compound shows maximum covalent character ?
(A) FeCl_2 (B) SnCl_2 (C) AlCl_3 (D) MgCl_2

52. The structure of IF_7 is
(A) Square pyramid (B) Trigonal pyramid
(C) Octahedral (D) Pentagonal bipyramid

53. The reduction of highly reactive metals is done by which of the following methods
(A) Calcination (B) Roasting
(C) Oxidation (D) Electrolytic refining

54. Which among the following is NOT an example of covalent bond ?
(A) LiF (B) NH_3 (C) CF_4 (D) HF

55. Which of the following does not cause the permanent hardness in water ?
(A) Nitrates (B) Sulphates
(C) Chlorides (D) Bicarbonates.

56. Who discovered the element hydrogen ?
(A) Henry Cavendish (B) Antoine Lavoisier
(C) Mendeleev (D) Newton.

57. Amines behave as
(A) Lewis acid (B) Lewis base
(C) Aprotic acid (D) Neutral compound

58. The molecule having smallest bond angle is
(A) NCl_3 (B) AsCl_3 (C) SbCl_3 (D) PCl_3

59. The increasing order of ionic radii of the given isoelectronic species is
(A) Cl^- , Ca^{2+} , K^+ , S^{2-} (B) S^{2-} , Cl^- , Ca^{2+} , K^+
(C) Ca^{2+} , K^+ , Cl^- , S^{2-} (D) K^+ , S^{2-} , Ca^{2+} , Cl^-

60. The size of nucleus of an atom is measured in ?
(A) a.m.u. (B) Angstrom
(C) Fermi (D) Cm

61. The most commonly used bleaching agent is
(A) Alcohol (B) Carbondioxide
(C) Chlorine (D) Sodium chloride

109. If the integral $\int \frac{5 \tan x}{\tan x - 2} = x + a \ln |\sin x - 2 \cos x| + k$

then a is equal to

- (A) -1 (B) -2 (C) 1 (D) 2

110. A equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is

- (A) $x - 2y + 3z - 3 = 0$ (B) $x - 2y + 2z + 1 = 0$
(C) $x - 2y + 2z - 1 = 0$ (D) $x - 2y + 2z + 5 = 0$

111. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, the value of k is

- (A) 29/5 (B) 5 (C) 6 (D) 11/5

112. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is

- (A) $20\sqrt{2}$ (B) $\frac{10\sqrt{2}}{3}$
(C) $\frac{20\sqrt{2}}{3}$ (D) $10\sqrt{2}$

113. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If two marbles are picked at random, what is the probability that both are red ?

- (A) $\frac{1}{7}$ (B) $\frac{1}{3}$ (C) $\frac{2}{15}$ (D) $\frac{2}{5}$

114. The area of triangle whose vertices are (3, 8), (-4, 2) and (5, 1), is (in sq. units)

- (A) $\frac{51}{2}$ (B) $\frac{61}{2}$ (C) 25 (D) 0

115. Let A be a non-singular matrix of order 3×3 . Then the value of $|\text{adj } A|$ is equal to

- (A) $|A|$ (B) $|A|^2$ (C) $|A|^3$ (D) $3|A|$

116. If $x = at^2$ and $y = 2at$, then the value of dy/dx

- (A) t (B) $\frac{1}{t^2}$ (C) $\frac{1}{t}$ (D) t^2

117. The value of the integral $\int_{-1}^1 \sin^5 x \cos^4 x \, dx$ is

- (A) 0 (B) 1 (C) $\frac{\pi}{2}$ (D) 2π

118. The point at which the tangent to the curve

$y = \sqrt{4x - 3} - 1$ has its slope equal to $\frac{2}{3}$

- (A) (3, 2) (B) (2, 3)
(C) (5, 4) (D) (0, 0)

119. The general solution of the differential equation

$$x \frac{dy}{dx} + 2y = x^2 \quad (x \neq 0) \text{ is}$$

- (A) $2x^2 + cy$ (B) $x^2 + cx$
(C) $\frac{x^2}{4} + cx^{-2}$ (D) $x^2 + 4x + 1$

120. The area of parallelogram whose adjacent sides are given

by vectors $\vec{a} = 3\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ is

- (A) $\sqrt{21}/2$ (B) $\sqrt{45}$ (C) $\sqrt{21}$ (D) $\sqrt{42}$

121. The direction cosines of line which makes angle 90° , 60° and 30° with positive direction of x , y and z -axis respectively, are

- (A) $\frac{-2}{3}, \frac{-1}{3}, \frac{2}{3}$ (B) $\frac{-1}{3}, \frac{2}{3}, \frac{-2}{3}$
(C) $\frac{2}{3}, \frac{1}{3}, \frac{-1}{3}$ (D) $0, \frac{1}{2}, \frac{\sqrt{3}}{2}$

122. The square roots of -3 are

- (A) $\pm\sqrt{3}$ (B) $\pm\sqrt{-3}$
(C) $\pm\sqrt{3}i$ (D) can't be determined.

123. The eccentricity and length of latus-rectum for the given type of ellipse : $\frac{x^2}{9} + y^2 = 1$, are

- (A) $2\sqrt{2}$ (B) $\left(\frac{\sqrt{3}}{2}\right)$ and 5
(C) $\frac{2\sqrt{2}}{3}$ and $\frac{2}{3}$ (D) None of these.

124. If α, β, γ are the roots of $2x^3 - 3x^2 + 6x + 1 = 0$, then $\alpha^2 + \beta^2 + \gamma^2$ is

- (A) $\frac{15}{4}$ (B) -3 (C) $-\frac{15}{4}$ (D) $\frac{33}{4}$

125. If the equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is (2, -1), then the length of its side is

- (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{\frac{3}{2}}$
(C) $\sqrt{\frac{2}{3}}$ (D) None of these.

126. The value of $\lim_{x \rightarrow \pi} \frac{\sin(\pi - x)}{\pi(\pi - x)}$ is

- (A) $\frac{2}{\pi}$ (B) $\frac{1}{\pi}$ (C) 2π (D) $\frac{3}{\pi}$

127. How many chords can be drawn through 21 points on a circle ?

- (A) 220 (B) 250 (C) 2 (D) 210

128. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of exactly 3 girls.

- (A) 502 (B) 500 (C) 501 (D) 504.

129. The absolute maximum and minimum value of the functions

$$f(x) = x^3, x \in [-2, 2] \text{ are}$$

- (A) (0, 8) (B) (8, -8)
(C) (2, -2) (D) (-8, 0)

130. What will be the probability of getting 53 Sundays in a non leap year ?

- (A) $\frac{2}{7}$ (B) $\frac{1}{7}$ (C) $\frac{6}{7}$ (D) $\frac{4}{7}$

131. The value of $\lim_{x \rightarrow 0} \frac{\log x}{\cot x}$ is
 (A) 0 (B) 1
 (C) ∞ (D) does not exist
132. The derivative of $y = x^{\sin x}$ is
 (A) $\cos x \cdot x^{\sin x - 1}$ (B) $\frac{\sin 2x}{2} x^{\sin x - 1}$
 (C) $x^{\sin x} \left(\cos x \log_e x + \frac{\sin x}{x} \right)$
 (D) $\cos x \cdot \log_e x + \frac{\sin x}{x}$
133. In determinant of matrix A is zero, then
 (A) A is a singular matrix
 (B) A is non-singular matrix
 (C) Can't say
 (D) Skew symmetric matrix.
134. Value of maxima of $\left(\frac{1}{x}\right)^x$ is
 (A) e (B) $e^{(1/e)}$
 (C) $(1/e)^e$ (D) e^e
135. The number of permutations of the letters of the word ALLAHABAD, are
 (A) 7560 (B) 9650
 (C) 7520 (D) 752
136. For what value of n , ${}^nP_5 = 42 {}^nP_3$, given that $n > 4$
 (A) -3 (B) 4
 (C) 10 (D) 1
137. The value of $\cos(-1710^\circ)$ is
 (A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) 0
138. The value of $\sin 15^\circ$ is
 (A) $\frac{2\sqrt{2}}{\sqrt{3}-1}$ (B) $\frac{2\sqrt{2}}{\sqrt{3}+1}$
 (C) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
139. For what value of a , the 17th and 18th terms of expansion $(2+a)^{50}$ are equal
 (A) 2 (B) 1
 (C) 0 (D) -2
140. The coefficient of x^6y^3 in the expansion $(x+2y)^9$ is
 (A) 670 (B) 671
 (C) 672 (D) None of these.
141. What does DMA stands for?
 (A) Distinct Memory Access
 (B) Direct Memory Allocation
 (C) Direct Module Access
 (D) Direct Memory Access.
142. ASCII stands for
 (A) American Stable Code for International Interchange.
 (B) American Standard Case for International Interchange.
 (C) American Standard Code for Information Interchange.
 (D) American Standard Code for Interchange Information.
143. Who invented the high level C language?
 (A) Dennis M. Ritchie (B) Nikalus Writh
 (C) Seymour Papert (D) Donald Kunth.
144. To prevent the loss of data during power failure, use $a(n)$:
 (A) Encryption program (B) Surge protector
 (C) Firewall (D) UPS
145. Physical or logical arrangement of network is called
 (A) Topology (B) Routing
 (C) Networking (D) None of these.
146. WAN stands for
 (A) World Area Network (B) Wide Area Network
 (C) Web Area Network (D) None of these.
147. Which one of the following is a transport layer protocol used in networking?
 (A) TCP (B) UDP
 (C) Both TCP and UDP (D) None of these.
148. Amazon is a best example of site
 (A) Blogging (B) E-commerce
 (C) Social Networking (D) Entertainment
149. URI is called as
 (A) Web address (B) System software
 (C) Hardware (D) Operating system
150. The example of personal computer OS are :
 (A) DOS (B) Mac DOS
 (C) Windows (D) All of these.

ANSWER

- | | | | | |
|----------|----------|----------|----------|----------|
| 1. (C) | 2. (C) | 3. (B) | 4. (A) | 5. (A) |
| 6. (B) | 7. (D) | 8. (B) | 9. (B) | 10. (A) |
| 11. (B) | 12. (D) | 13. (D) | 14. (C) | 15. (C) |
| 16. (D) | 17. (A) | 18. (C) | 19. (C) | 20. (B) |
| 21. (A) | 22. (C) | 23. (D) | 24. (D) | 25. (A) |
| 26. (B) | 27. (A) | 28. (B) | 29. (B) | 30. (B) |
| 31. (B) | 32. (B) | 33. (B) | 34. (C) | 35. (A) |
| 36. (B) | 37. (D) | 38. (D) | 39. (C) | 40. (C) |
| 41. (B) | 42. (B) | 43. (C) | 44. (C) | 45. (A) |
| 46. (D) | 47. (C) | 48. (D) | 49. (D) | 50. (B) |
| 51. (C) | 52. (D) | 53. (D) | 54. (A) | 55. (D) |
| 56. (A) | 57. (B) | 58. (C) | 59. (C) | 60. (B) |
| 61. (C) | 62. (B) | 63. (D) | 64. (B) | 65. (B) |
| 66. (B) | 67. (C) | 68. (B) | 69. (C) | 70. (C) |
| 71. (D) | 72. (B) | 73. (D) | 74. (C) | 75. (C) |
| 76. (C) | 77. (B) | 78. (D) | 79. (C) | 80. (A) |
| 81. (B) | 82. (A) | 83. (D) | 84. (C) | 85. (B) |
| 86. (B) | 87. (C) | 88. (D) | 89. (B) | 90. (B) |
| 91. (C) | 92. (A) | 93. (C) | 94. (D) | 95. (D) |
| 96. (A) | 97. (A) | 98. (B) | 99. (B) | 100. (C) |
| 101. (B) | 102. (C) | 103. (D) | 104. (B) | 105. (A) |
| 106. (D) | 107. (D) | 108. (C) | 109. (D) | 110. (A) |
| 111. (C) | 112. (C) | 113. (A) | 114. (B) | 115. (B) |
| 116. (C) | 117. (A) | 118. (A) | 119. (C) | 120. (D) |
| 121. (D) | 122. (C) | 123. (C) | 124. (C) | 125. (C) |
| 126. (B) | 127. (D) | 128. (D) | 129. (B) | 130. (B) |
| 131. (A) | 132. (C) | 133. (A) | 134. (B) | 135. (A) |
| 136. (C) | 137. (D) | 138. (D) | 139. (B) | 140. (C) |
| 141. (D) | 142. (C) | 143. (A) | 144. (D) | 145. (A) |
| 146. (B) | 147. (C) | 148. (B) | 149. (A) | 150. (D) |